

Tian Lan

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The Chinese University of Hong Kong
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Education

- 2013–2017 PhD in Theoretical Physics
Perimeter Institute for Theoretical Physics & University of Waterloo
Thesis: [A Classification of \(2+1\)D Topological Phases with Symmetries](#)
Supervisor: Xiao-Gang Wen
- 2012–2013 MSc in Theoretical Physics, Perimeter Scholars International (PSI)
Perimeter Institute for Theoretical Physics & University of Waterloo
- 2008–2012 BSc in Mathematics and Physics, Tsinghua University

Experience

- 2018–2020 Postdoctoral Fellow
Institute for Quantum Computing, University of Waterloo
Supervisors: Raymond Laflamme, Bei Zeng
- 2021.6–now Assistant Professor
The Chinese University of Hong Kong

Research Interests

My research focuses on the theory of topological phases of matter:

- intrinsic topological order,
- symmetry protected topological (SPT) phases, which include topological insulators as a special case,
- symmetry enriched topological (SET) phases,
- higher dimensional generalization of symmetry and topological phases,
- ...

and their mathematical foundation:

- tensor category,

- higher category, higher algebra,
- group cohomology, algebraic topology,
- ...

I am also interested in understanding the nature of phases of matter from a quantum information perspective, and possible applications of topological materials such as quantum computing.

Grants

PI, Early Career Scheme (ECS) No. 24304722, Hong Kong Research Grants Council

PI, Guangdong Provincial Quantum Science Strategic Initiative No. GDZX2501013

Co-PI, Collaborative Research Fund (CRF) No. C7015-24GF, Hong Kong Research Grants Council

Teaching

2021-25 Fall PHYS5410 Advanced Quantum Mechanics

2024-26 Spring PHYS1111 University Physics I

Referee

Physical Review Series (PRX, PRX Quantum, PRL, PRRResearch, PRB)

Communications in Mathematical Physics, Journal of High Energy Physics, SciPost Physics, Quantum ...

Associate Editor

Advances in Theoretical and Mathematical Physics.

Talks

- A “Periodic Table” of Topological Orders
 - SIAM Annual Meeting (AN16), Boston, Massachusetts, USA, July, 2016 (invited but was not able to attend due to visa delay)
 - Department of Physics, Fudan University, May, 2018
 - Department of Physics, Tsinghua University, July, 2018
 - Graduate School of China Academy of Engineering Physics, July, 2018
 - Department of Physics, Southern University of Science and Technology, July, 2019
 - The Chinese University of Hong Kong, October, 2020
- Non-Abelian Family of Topological Order²¹
 - Croucher Summer Course “Quantum Entanglement and Topological Order”, The Chinese University of Hong Kong, June, 2017
 - Department of Physics, Southern University of Science and Technology, July, 2019
- Matrix Formulation of Non-Abelian Family¹³
 - Topological Quantum Computing (TQC2019), Shenzhen, China, December, 2019
- Higher Dimensional Topological Order, Higher Category and A Classification in $3+1D$ ^{14,17}

- Department of Physics, Tsinghua University, May, 2018
 - Séminaire de mathématiques supérieures (SMS) 2018, Fields Institute for Research in Mathematical Sciences, June, 2018
 - Quantum Information and Operator Algebras III, Academy of Mathematics and Systems Science, Chinese Academy of Sciences, July, 2018
 - Department of Physics, Sun Yat-Sen University, July, 2019
 - QuIST V: The 5th International Conference on Quantum Information, Space-time, and Topological order, Kunming, China, August, 2019
 - Department of Physics, Southern University of Science and Technology, August, 2019
 - The Chinese University of Hong Kong, October, 2020
 - CMSA, Harvard University, June, 2021
 - BIMS, March, 2022
- The Modular Extension Characterization of SPT/SET Phases^{19,20,23}
 - BIMS, December, 2021
 - Introduction to Categorical Approach to Topological Phases in Arbitrary Dimensions
 - Center of Mathematical Sciences and Applications, Harvard University, April 15, 2020
 - Higher symmetry enriched topological phases^{10,11}
 - Topological Orders and Higher Structures, Erwin Schrödinger Institute for Mathematics and Physics, University of Vienna, August, 2020
 - Quantum Current⁷
 - Croucher Summer Course "Quantum Entanglement and Topological Order", CUHK, June, 2023
 - The first International Congress of Basic Science (ICBS), BIMS, July, 2023
 - Entangled Quantum Matter Workshop, Tsinghua University, August, 2023
 - Gauging of generalized symmetry/Category of SET orders⁶
 - Mini-Workshop on Unconventional Superconductivity and Correlated Electron Systems, CUHK, Dec, 2023
 - The International Congress of Basic Science (ICBS), BIMS, July, 2024
 - Applications of Generalized Symmetries and Topological Defects to Quantum Matter, Simons Center for Geometry and Physics, September, 2024
 - Tube category, tensor renormalization and topological holography (Origin of topological holography)²
 - Tensor Network States: Algorithms and Applications, CUHK, December, 2024
 - The University of Hong Kong, June, 2025
 - SUSTech Summer School on Generalized Symmetry and Quantum Criticality, October, 2025
 - MIST Conference, CUHK, February, 2026

Publications

- [1] Gen Yue, Ansi Bai, Linqian Wu, and **Tian Lan**, “Pro-tensor network”, (2026), [arXiv:2605.06661 \[cond-mat.str-el\]](#).
- [2] **Tian Lan**, “Tube category, tensor renormalization and topological holography”, [Communications in Mathematical Physics](#) **406**, 196 (2025), [arXiv:2412.07198 \[math-ph\]](#).
- [3] Wenjie Xi, **Tian Lan**, Longye Wang, Chenjie Wang, and Wei-Qiang Chen, “On a class of fusion 2-category symmetry: condensation completion of braided fusion category”, [Advances in Theoretical and Mathematical Physics](#) **29**, 151–204 (2025), [arXiv:2312.15947](#).
- [4] Gen Yue, Longye Wang, and **Tian Lan**, “Condensation completion and defects in 2+1d topological orders”, [Physical Review B](#) **112**, 104106 (2025), [arXiv:2402.19253 \[cond-mat.str-el\]](#).
- [5] Ying Chan, **Tian Lan**, and Linqian Wu, “Torus algebra and logical operators at low energy”, [10.48550/ARXIV.2403.01577](#) (2024), [arXiv:2403.01577](#).
- [6] **Tian Lan**, Gen Yue, and Longye Wang, “Category of SET orders”, [Journal of High Energy Physics](#) **2024**, 111 (2024), [arXiv:2312.15958](#).
- [7] **Tian Lan** and Jing-Ren Zhou, “Quantum current and holographic categorical symmetry”, [SciPost Physics](#) **16**, 053 (2024), [arXiv:2305.12917](#).

- [8] Chenqi Meng, Xinping Yang, **Tian Lan**, and Zhengcheng Gu, “Non-invertible spts: an on-site realization of $(1+1)d$ anomaly-free fusion category symmetry”, [10.48550/ARXIV.2412.20546](#) (2024), [arXiv:2412.20546 \[cond-mat.str-el\]](#).
- [9] Wenjie Xi, Ya-Lei Lu, **Tian Lan**, and Wei-Qiang Chen, “A lattice realization of general three-dimensional topological order”, [10.48550/ARXIV.2110.06079](#) (2021), [arXiv:2110.06079](#).
- [10] Liang Kong, **Tian Lan**, Xiao-Gang Wen, Zhi-Hao Zhang, and Hao Zheng, “Algebraic higher symmetry and categorical symmetry: a holographic and entanglement view of symmetry”, *Physical Review Research* **2**, 043086 (2020), [arXiv:2005.14178](#).
- [11] Liang Kong, **Tian Lan**, Xiao-Gang Wen, Zhi-Hao Zhang, and Hao Zheng, “Classification of topological phases with finite internal symmetries in all dimensions”, *Journal of High Energy Physics* **2020**, 93 (2020), [arXiv:2003.08898](#).
- [12] **Tian Lan**, Xueda Wen, Liang Kong, and Xiao-Gang Wen, “Gapped domain walls between 2+1D topologically ordered states”, *Physical Review Research* **2**, 023331 (2020), [arXiv:1911.08470](#).
- [13] **Tian Lan**, “Matrix formulation for non-Abelian families”, *Physical Review B* **100**, 241102(R) (2019), [arXiv:1908.02599](#).
- [14] **Tian Lan** and Xiao-Gang Wen, “Classification of 3+1D bosonic topological orders (II): the case when some pointlike excitations are fermions”, *Physical Review X* **9**, 021005 (2019), [arXiv:1801.08530](#).
- [15] **Tian Lan**, Chenchang Zhu, and Xiao-Gang Wen, “Fermion decoration construction of symmetry-protected trivial order for fermion systems with any symmetry and in any dimension”, *Physical Review B* **100**, 235141 (2019), [arXiv:1809.01112](#).
- [16] Chenchang Zhu, **Tian Lan**, and Xiao-Gang Wen, “Topological nonlinear σ -model, higher gauge theory, and a systematic construction of 3+1D topological orders for boson systems”, *Physical Review B* **100**, 045105 (2019), [arXiv:1808.09394](#).
- [17] **Tian Lan**, Liang Kong, and Xiao-Gang Wen, “Classification of 3+1D bosonic topological orders: the case when pointlike excitations are all bosons”, *Physical Review X* **8**, 021074 (2018), [arXiv:1704.04221](#).
- [18] **Tian Lan**, “A classification of $(2+1)D$ topological phases with symmetries”, PhD thesis (University of Waterloo, Sept. 2017), p. 86, [arXiv:1801.01210](#).
- [19] **Tian Lan**, Liang Kong, and Xiao-Gang Wen, “Classification of $(2+1)$ -dimensional topological order and symmetry-protected topological order for bosonic and fermionic systems with on-site symmetries”, *Physical Review B* **95**, 235140 (2017), [arXiv:1602.05946](#).
- [20] **Tian Lan**, Liang Kong, and Xiao-Gang Wen, “Modular extensions of unitary braided fusion categories and 2+1D topological/SPT orders with symmetries”, *Communications in Mathematical Physics* **351**, 709–739 (2017), [arXiv:1602.05936](#).
- [21] **Tian Lan** and Xiao-Gang Wen, “Hierarchy construction and non-Abelian families of generic topological orders”, *Physical Review Letters* **119**, 040403 (2017), [arXiv:1701.07820](#).
- [22] Keren Li, Yidun Wan, Ling-Yan Hung, **Tian Lan**, Guilu Long, Dawei Lu, Bei Zeng, and Raymond Laflamme, “Experimental identification of non-Abelian topological orders on a quantum simulator”, *Physical Review Letters* **118**, 080502 (2017), [arXiv:1608.06932](#).
- [23] **Tian Lan**, Liang Kong, and Xiao-Gang Wen, “Theory of $(2+1)$ -dimensional fermionic topological orders and fermionic/bosonic topological orders with symmetries”, *Physical Review B* **94**, 155113 (2016), [arXiv:1507.04673](#).
- [24] **Tian Lan**, Juven C. Wang, and Xiao-Gang Wen, “Gapped domain walls, gapped boundaries, and topological degeneracy”, *Physical Review Letters* **114**, 076402 (2015), [arXiv:1408.6514](#).
- [25] **Tian Lan** and Xiao-Gang Wen, “Topological quasiparticles and the holographic bulk-edge relation in $(2+1)$ -dimensional string-net models”, *Physical Review B* **90**, 115119 (2014), [arXiv:1311.1784](#).